

Challenge Python Engineer

With the CSV attached, we ask you to implement a solution for sentiment classification of movie reviews. The solution should only consider the Dutch language reviews and should classify those reviews as either: Positive, Average, or Negative. Create a small app to use this solution (ideally a ML model) with proper instructions on how to run it. You are free to report the results in a way that you think is suitable for the task. Below you can find some pointers but feel free to improvise and create your own app in your own way.

- 1) Hand-craft or automatically create features that you think would be useful to classify the reviews.
- 2) Store the model training experiments in a way that it is easy to revisit and see/compare their metrics.
- 3) Follow proper software design principles and object oriented programming practices.
- 4) Makes sure to include at least one unittest for one of your classes (preferably your model class).
- 5) For inference purposes, create a REST API with a "/classify" endpoint that would take a single review (i.e. corresponding to a row in the dataset), and perform inference (e.g., required data transformations and model prediction) and finally return the label: Positive, Average, or Negative.. A bonus point is if you also serve this API inside a Docker container.
- 6) Pay attention to latency on inference time on top of performance metrics.
- 7) Bonus: Be able to explain why the model makes its predictions.
- 8) Bonus: Version your data and model properly.

We are looking to discuss more your approach to the problem rather than evaluating the model's final accuracy. Best of luck!